

Day 1 – C++ Practice Programs

Introduction to C++ | Arrays, Vectors & Strings

Program 1: Move Zeros to Last Index

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {0, 8, 0, 7, 13};
    int j = 0;

    for (int i = 0; i <= 4; i++) {
        if (arr[i] != 0) {
            arr[j] = arr[i];
            j++;
        }
    }

    while (j <= 4) {
        arr[j] = 0;
        j++;
    }

    for (int i = 0; i <= 4; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

Program 2: Count Number of Zeros in a Number

```
#include <iostream>
using namespace std;

int main() {
    int n = 2010;
    int count = 0;

    while (n != 0) {
        int last = n % 10;
        if (last == 0) {
            count++;
        }
        n = n / 10;
    }

    cout << count;
    return 0;
}
```

```
}
```

Program 3: Increment / Decrement & Compound Assignment Trace

This page in the notebook is a hand-traced 'predict the output' exercise on increment/decrement and compound assignment operators. Comments show the expected value at each step.

```
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    int x = 25;

    x *= 2;           // x = 50
    cout << -x + 2;    // -48
    a += 24;          // a = 34
    cout << --a;       // 33
    a %= x;           // 33 % 50 = 33
    cout << x % 10;    // 0
    cout << a++;        // 33 (a becomes 34)
    --x;              // x = 49
    a--;              // a = 33
    cout << a++;        // 33 (a becomes 34)
    cout << --x;        // 48

    int b = 10;
    a /= b;           // a = 34 / 10 = 3
    cout << ++b;        // 11

    return 0;
}
```

Program 4: Check Leap Year

```
#include <iostream>
using namespace std;

int main() {
    int year;
    cin >> year;

    if (year % 100 == 0) {
        if (year % 400 == 0) {
            cout << "Leap year";
        } else {
            cout << "Not a leap year";
        }
    }
    else if (year % 4 == 0) {
        cout << "Leap year";
    }
    else {
```

```

        cout << "Not a leap year";
    }

    return 0;
}

```

Vector vs Array (Theory)

A **vector** is a dynamic array provided by the Standard Template Library (STL). Unlike a normal array, a vector can **grow or shrink automatically** in size.

Program 5: Vector Basics – push_back, at, pop_back, size

```

#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> numbers;
    numbers.push_back(3);
    numbers.push_back(7);
    numbers.push_back(31);

    cout << numbers.at(1) << endl;

    for (int x : numbers) {
        cout << x << " ";
    }

    cout << " " << endl;

    numbers.pop_back();
    int n = numbers.size();

    for (int x : numbers) {
        cout << x << " ";
    }

    cout << endl;
    cout << numbers.at(0);

    return 0;
}

```

Program 6: Vector Using Even or Odd Number

Only the setup (vector declaration) for this program was written on the page; the even/odd classification logic was not completed in the notes.

```

#include <iostream>
#include <vector>
using namespace std;

```

```

int main() {
    vector<int> nums = {10, 21, 11, 14, 19};

    // (Program continues in the notebook - to classify
    //  each element as even or odd)

    return 0;
}

```

Program 7: Find the Second Largest Element

```

#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> nums = {10, 21, 11, 14, 19};
    int f = 0;    // largest
    int s = 0;    // second largest
    int n = nums.size();

    for (int i = 0; i < n; i++) {
        if (nums[i] > f) {
            s = f;
            f = nums[i];
        }
        else if (nums[i] != f && nums[i] > s) {
            s = nums[i];
        }
    }

    cout << s;
    return 0;
}

```

Strings (Theory)

A **string** is a sequence of characters used to store text.

Program 8: String Input – cin vs getline

```

#include <iostream>
#include <string>
using namespace std;

int main() {
    string s;
    cin >> s;
    cout << s << endl;

    string s1;
    getline(cin, s1);
}

```

```
    cout << s1;

    return 0;
}
```

String Methods (Reference)

1. s.length()
2. s.size()
3. s.empty()
4. s.append()
5. s.push_back()
6. s.substr() – extract part of the string
7. s.find()
8. s.clear()

Program 9: To Find the Length of the Last Word

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string s = "The Goalsds";    // index: 0123456789

    int i = s.length() - 1;
    while (s[i] == ' ') {
        i--;
    }

    int len = 0;
    while (i >= 0 && s[i] != ' ') {
        len++;
        i--;
    }

    cout << len;
    return 0;
}
```